

Our first approach: structured regex rules

WHY WE STARTED HERE

The team first tested a rule-based approach because it offered a controlled and transparent way to automate recurring comment patterns.

- No external AI dependency**
Runs locally without an API call or per-comment AI cost.
- Consistent results**
The same comment produces the same result every time.
- Fast processing**
A fixed set of compiled patterns runs quickly across many rows.
- Traceable decisions**
Each extracted fact links to the pattern and matched source text.

HOW THE APPROACH WORKED

- 1 Pattern library**
About 25 patterns targeted rates, dates, age limits, authorization, codes, service limits, methodology terms, and change language.
- 2 Fact extraction**
Each match captured the field, value, source text, pattern name, and confidence level.
- 3 Rule comparison**
The extracted facts were checked against CSRA conditions, SOP rule text, or available row data.
- 4 Decision output**
The engine returned no change, new rule, update, or conflict with suggested changes.
- 5 Review package**
Batch results and draft SQL were generated for human review. Nothing executed automatically.

BUSINESS CONSTRAINTS WE DISCOVERED

Wording variation

A new phrase required another pattern to be written and tested.

Limited context

The rules could find a value but could not always understand how the full comment changed its meaning.

Ongoing maintenance

New states, authors, and rule types continuously expanded the pattern library.

Why we shifted to Azure OpenAI

The business needed a solution that could understand varied analyst language without adding a new rule for every new phrase.

BUSINESS IMPACT		
BEFORE	NOW	RESULT
Hours or days of manual review	Minutes to process the same work	Consistent decisions with human oversight

THE BUSINESS REASON FOR THE SHIFT

- Understands varied wording**
The language model interprets comments written in different styles while preserving the intended rule change.
- Uses the full context**
It evaluates the complete comment and compares it with the current rule before deciding what should happen.
- Faster to extend**
The team can update the instructions for new tasks instead of building and maintaining another set of phrase-specific patterns.

CONTROLLED BUSINESS PROCESS

- NO ACTION**
The current rule is already correct.
- UPDATE**
The solution identifies exactly what needs to change.
- HUMAN REVIEW**
The confidence is too low, so a reviewer receives the case.

SECURITY AND OVERSIGHT

- 85% threshold**
An automatic change can proceed only when confidence reaches the defined threshold.
- Human control**
Anything below the threshold goes to a reviewer instead of being guessed.
- Full visibility**
Every decision includes an explanation, is logged, and appears in the summary email.